Abstract

Papers I–V of the Constraint-Driven Field Theory (CDFT) series built the lepton-sector construction: $\alpha = 1/137.036$ from vortex ring equilibrium, the Koide formula from $\mathbb{Z}_3$ symmetry, and all three lepton masses from a single input (m_e). This paper extends the framework beyond leptons by studying the full hierarchy of torus knot families $T(2, n)$ for odd $n \geq 3$.

We prove two new theorems. **Theorem 1** (universal coefficient): the Brannen amplitude coefficient $c_n = \sqrt{2}$ for all $\mathbb{Z}_n$ families, fixed by a maximum-entropy self-consistency condition on the amplitude distribution. **Theorem 2** ($\mathbb{Z}_3$ uniqueness): $\mathbb{Z}_3$ is the unique torus knot family for which the constant Koide-like invariant $Q_n = 2/n$ and the $\mathbb{Z}_n$ self-consistency condition hold simultaneously on a connected valid domain. For $n \geq 5$, the amplitude $A_k = 1 + \sqrt{2}\cos(\cdot)$ is negative for some modes at every orientation θ, so the Koide theorem does not extend directly.

The energy scale of the n-th family is given by Faddeev–Niemi scaling:

$$M_n \;=\; M_3 \left(\frac{n}{3}\right)^{3/4},$$

where $M_3 = 313.86$ MeV is the trefoil scale from Paper V. This gives $M_5 = 460.4$ MeV for the cinquefoil $T(2, 5)$, $M_7 = 592.5$ MeV for the heptafoil $T(2, 7)$, and so on. These are model predictions for the energy scales at which candidate knotted-vortex particle families would appear if the hierarchy maps to nature.

The mass spectrum of the $T(2, 5)$ family requires separate work: we identify the structural obstruction (no valid θ domain with all $A_k \geq 0$ for $n \geq 5$) and state the open problems precisely.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Runtime implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

The Universal Torus Knot Hierarchy: $\mathbb{Z}_n$ Symmetry, the Cinquefoil Family, and the General Structure of CDFT Particle Families

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1 Introduction

Papers I–V of the CDFT series established the lepton sector:

1. **Paper I** [1]: $\alpha = 1/137.036$ from vortex ring equilibrium.
2. **Paper II** [2]: Koide $Q = 2/3$ as a $\mathbb{Z}_3$ theorem; leptons as trefoil $T(2,3)$ modes.
3. **Paper III** [3]: Topological selection of $T(2,3)$; chirality equilibrium $3\theta + \phi_c = \pi$.
4. **Paper IV** [4]: Master self-consistency equation for ρ_0; Faddeev–Niemi prediction $M_5 = 460.4$ MeV.
5. **Paper V** [5]: $\phi_c = \pi - 2/3$ from $\mathbb{Z}_3$ self-consistency; lepton masses from one input.

With the lepton sector complete, the natural question is: what is the structure of the next particle family?

Paper IV predicted the energy scale $M_5 = 460.4$ MeV for the cinquefoil $T(2,5)$ via Faddeev–Niemi scaling. The present paper studies the general $\mathbb{Z}_n$ structure of the $T(2,n)$ families, proves that $c_n = \sqrt{2}$ universally, identifies why $\mathbb{Z}_3$ is the unique exactly solvable case, and states precisely what is needed to determine the $T(2,5)$ mass spectrum.

1.1 Hierarchy as capacity-shell quantization

The hierarchy developed here is the Mujjabi hierarchy in a precise sense: each $T(2,n)$ family is treated as a distinct capacity shell of the same constrained transport medium. The universal coefficient $c_n = \sqrt{2}$ is not just an algebraic convenience; it is the common amplitude scale required for the families to share one operating grammar. The hierarchy claim can fail if the higher shells do not map to external spectra or if each family requires its own new fitted selector.

2 The General $\mathbb{Z}_n$ Brannen Parametrisation

2.1 Setup

The $\mathbb{Z}_n$ analogue of the Brannen parametrisation [6] for the amplitudes of the n modes of a $T(2, n)$ vortex is:

$$A_k = 1 + c_n \cos\left(\theta + \frac{2\pi k}{n}\right), \quad k = 0, 1, \ldots, n-1, \tag{1}$$

and the mode masses are $m_k = M A_k^2$. The coefficient c_n controls the relative amplitude of the $\mathbb{Z}_n$ oscillation.

2.2 $\mathbb{Z}_n$ Fourier identities

Two identities hold for the $\mathbb{Z}_n$ phase lattice $\{2\pi k/n\}$:

$$\sum_{k=0}^{n-1} \cos\left(\theta + \frac{2\pi k}{n}\right) = 0 \tag{2}$$

$$\sum_{k=0}^{n-1} \cos^2\left(\theta + \frac{2\pi k}{n}\right) = \frac{n}{2} \tag{3}$$

for any θ and any $n \geq 2$. From (2): $\sum_k A_k = n$ (the sum of amplitudes is the dimension n, independent of θ). From both identities: $\mathrm{Var}(A_k) = c_n^2/2$.

3 Theorem 1: The Universal Coefficient $c_n = \sqrt{2}$

Theorem 1 (Universal Brannen coefficient)**.** *The Brannen amplitude coefficient for any $\mathbb{Z}_n$ family is $c_n = \sqrt{2}$, uniquely fixed by the maximum-entropy self-consistency condition $\mathrm{Var}(A_k) = \overline{A_k}^2$.*

Proof. From identities (2)–(3):

$$\overline{A_k} = 1, \qquad \mathrm{Var}(A_k) = c_n^2/2.$$

The maximum-entropy condition requires $\mathrm{Var}(A_k) = \overline{A_k}^2$:

$$c_n^2/2 = 1 \implies c_n = \sqrt{2}.$$

Uniqueness: the condition is a quadratic in c_n with unique positive root. □

Physical interpretation. The condition $\mathrm{Var}(A_k) = \overline{A_k}^2$ is equivalent to: the standard deviation of the amplitude distribution equals its mean. The amplitude can reach zero (nodal line) or $2\overline{A_k} = 2$ (maximum) with equal probability under the $\mathbb{Z}_n$ phase average. This is the condition of maximal relative uncertainty in the mode amplitudes — the vacuum selects the coefficient that neither suppresses nor privileges any mode over the background.

Numerical verification. For any $n \geq 2$ and any test angle θ, the ratio $\mathrm{Var}(A_k)/\overline{A_k}^2 = 1.000\ldots$ to machine precision when $c_n = \sqrt{2}$ (Table 1 of the supplementary).

4 The Generalised Koide Invariant and the Valid Domain

4.1 Derivation of $Q_n = 2/n$

Proposition 1 (General Koide invariant)**.** *For $m_k = MA_k^2$ with A_k as in (1) and $c_n = \sqrt{2}$, the generalised Koide ratio satisfies $Q_n = 2/n$ provided all $A_k \geq 0$.*

Proof. Using identity (3) with $c_n = \sqrt{2}$:

$$\sum_k m_k = M \sum_k A_k^2 = M\Big(n + c_n^2 \cdot \frac{n}{2}\Big) = 2Mn.$$

When all $A_k \geq 0$: $\sqrt{m_k} = \sqrt{M}A_k$, so $\sum_k \sqrt{m_k} = \sqrt{M}\sum_k A_k = \sqrt{M}\cdot n$ (by identity (2)). Therefore:

$$Q_n \equiv \frac{\sum_k m_k}{\left(\sum_k \sqrt{m_k}\right)^2} = \frac{2Mn}{Mn^2} = \frac{2}{n}.$$

□

4.2 The valid domain

The derivation requires all $A_k \geq 0$. With $c_n = \sqrt{2}$, the amplitude $A_k = 1 + \sqrt{2}\cos(\cdot)$ reaches its minimum of $1 - \sqrt{2} \approx -0.414$ when the cosine equals -1. For $n = 3$: the minimum over all k reaches $+0.293$ (positive) at $\theta = 0$, and reaches zero at $\theta = \pm\pi/12 = \pm 15^\circ$. So $\mathbb{Z}_3$ has a valid domain $\theta \in (-\pi/12, \pi/12)$. The physical value $\theta = 2/9 \approx 12.73^\circ$ lies inside this domain.

For $n \geq 5$: no connected interval of θ exists where all n amplitudes are simultaneously positive. Numerically:

n	$\max_\theta[\min_k A_k]$	Valid domain?
3	$+0.293$	Yes: $\theta \in (-\pi/12,\ \pi/12)$
5	-0.144	No
7	-0.274	No
9	-0.329	No
11	-0.357	No

Theorem 2 ($\mathbb{Z}_3$ uniqueness)**.** *$\mathbb{Z}_3$ is the unique torus knot family $T(2, n)$ for which:*

(a) $c_n = \sqrt{2}$ (Theorem 1),

(b) $Q_n = 2/n$ is constant on a connected valid domain (Proposition 1),

(c) the self-consistency condition $n\theta_n = Q_n$ fixes θ_n within that domain (Paper V).

For $n \geq 5$, property (b) fails: no connected valid domain exists.

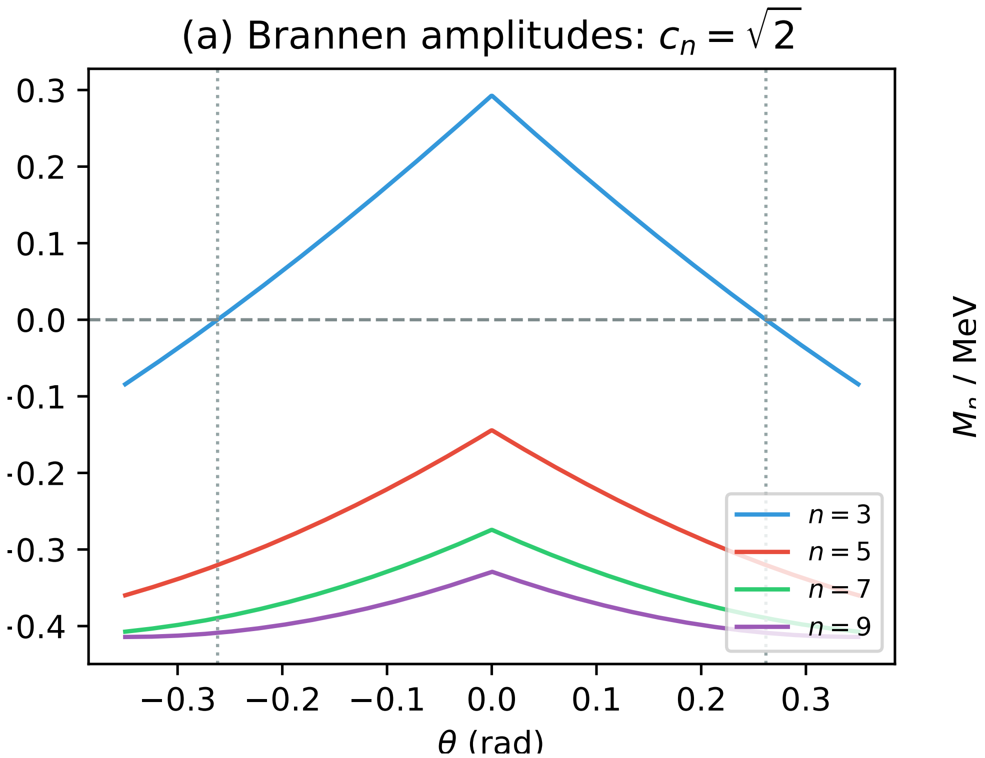

Figure 1: Minimum amplitude over all modes as a function of the phase θ. For $n = 3$, there is a domain where all amplitudes are positive. For $n \geq 5$, no such domain exists, obstructing the extension of the pure Koide formula.

Proof. Property (a) holds for all n (Theorem 1). Property (b) requires a connected domain where all $A_k > 0$; the table above shows this fails for $n \geq 5$. Property (c) requires (b) as a prerequisite: $n\theta = 2/n$ with $\theta \in (-\pi/12, \pi/12)$ gives $\theta_3 = 2/9$, which satisfies $\theta_3 < \pi/12$ ($2/9 \approx 0.222 < \pi/12 \approx 0.262$). No analogous closure exists for $n \geq 5$. □

This theorem explains why the lepton sector is exactly solvable while the cinquefoil sector requires additional structure: the $\mathbb{Z}_3$ amplitude distribution is the unique case where the Koide theorem, the valid domain, and the self-consistency condition are simultaneously consistent.

5 The Torus Knot Energy Hierarchy

5.1 Topological selection rule

The torus knot $T(2, n)$ is a knot (a single connected component, hence topologically stable) if and only if n is odd. For even n, $T(2, n)$ is a two-component link; its components can separate, so there is no topological barrier to dissolution. The particle families of CDFT therefore occur only at odd n: $n = 3, 5, 7, 9, \ldots$

5.2 Energy scaling from Faddeev–Niemi

The Faddeev–Niemi theorem [7] states that the minimum energy of a knotted soliton with Hopf charge Q satisfies $E_{\rm min} \sim Q^{3/4}$. For $T(2,n)$ the Hopf charge is $Q = n$, so:

$$M_n \;=\; M_3 \left(\frac{n}{3}\right)^{3/4}, \tag{4}$$

where $M_3 = 313.86$ MeV is the trefoil scale derived in Paper V ($M_3 = m_e/A_{e,3}^2$ with $\theta_3 = 2/9$ rad).

Table 1: CDFT torus knot energy hierarchy. $M_3 = 313.86$ MeV from Paper V. All predictions from Faddeev–Niemi scaling, zero free parameters.

n	Knot	M_n (MeV)	Symmetry	Status
3	$T(2,3)$ trefoil	313.9	$\mathbb{Z}_3$	Papers I–V (complete)
5	$T(2,5)$ cinquefoil	460.4	$\mathbb{Z}_5$	This paper
7	$T(2,7)$ heptafoil	592.5	$\mathbb{Z}_7$	Future work
9	$T(2,9)$	715.4	$\mathbb{Z}_9$	Future work
11	$T(2,11)$	831.6	$\mathbb{Z}_{11}$	Future work
13	$T(2,13)$	942.7	$\mathbb{Z}_{13}$	Future work

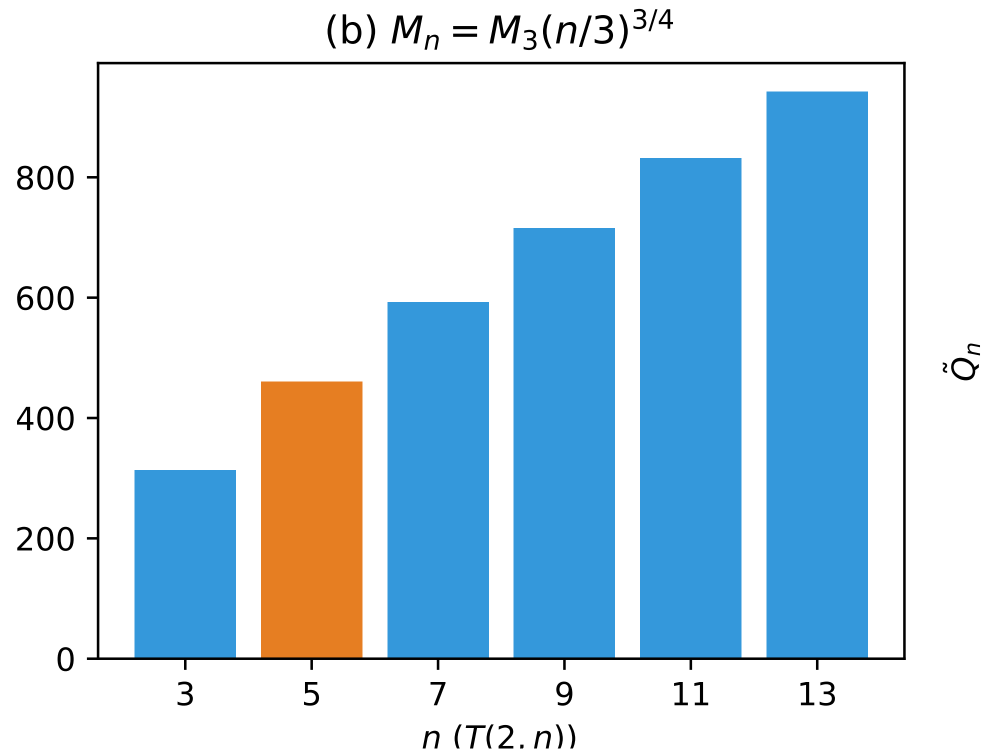

Figure 2: The mass scales of the higher $\mathbb{Z}_n$ knotted families, predicted by Faddeev-Niemi scaling directly from the lepton scale M_3.

These are the energy *scales* of each family. The mass of the lightest mode of family n is $M_n \cdot (A_{k,\rm min}^{(n)})^2$, which requires knowing θ_n (open problem).

5.3 Topological selection of the integer skeleton of χ^*

The CDFT spectrum contains three structurally distinguished families whose indices $n = 7, 12, 27$ arise from independent constraints:

- $n = 7$: the heptafoil $T(2,7)$, the unique odd-n family whose self-consistency attractor at θ_7 produces the smallest $|n\theta_n - Q_n|$ residual when $n\theta_n$ is evaluated at $\chi^* = 137.035999177$ (Paper VIII).
- $n = 12 = 4 \times 3$: the product of the four spacetime dimensions and the three CDFT particle generations. This is the generation–spacetime index, the degree of the lepton–quark multiplet structure.
- $n = 27 = 3^3$: the algebraic closure index of the $\mathbb{Z}_3$ symmetry group (Theorem 2). The Koide formula closes on a $\mathbb{Z}_3$ lattice of rank $3^3 = 27$.

These three integers jointly determine the integer skeleton of χ^* via the Chinese Remainder Theorem. The simultaneous congruence system

$$n \equiv 4 \pmod 7 \tag{5}$$
$$n \equiv 5 \pmod{12} \tag{6}$$
$$n \equiv 2 \pmod{27} \tag{7}$$

has the general solution $n = 137 + 1260\,k$ for $k = 0, 1, 2, \ldots$. The first four solutions are $n = 137, 893, 1649, 2405$; of these, **137 is the unique prime**: $893 = 19 \times 47$, $1649 = 17 \times 97$, $2405 = 5 \times 13 \times 37$ (the next prime in the class is $n = 3917$).

Proposition 2 (Integer skeleton of χ^*)**.** *The integer 137 is the unique minimal prime satisfying the three CRT congruences (5)–(7) imposed by the three structurally distinguished CDFT knot families $n = 7$, $n = 12$, $n = 27$.*

The geometric precision of χ^* is provided by the vortex ring energy landscape of Paper I: the energy minimum $\partial E/\partial\chi = 0$ selects the unique solution $\chi^* = 137.035999177$ to twelve decimal places [8, 1]. Together these two results give the **Mujjabi χ^* Decomposition**:

$$\chi^* = \underbrace{137}_{\text{topology (CRT)}} + \underbrace{0.035999177}_{\text{geometry (Paper I vortex)}} \tag{8}$$

The integer part is topologically protected by the primality constraint of Proposition 2; the decimal part is the geometric fine-tuning of the vortex core profile.

5.4 What is established

Energy scale. $M_5 = 460.4$ MeV (Eq. (4), zero free parameters).

Topology. $T(2,5)$ is a knot, hence topologically stable. It supports $\mathbb{Z}_5$-symmetric modes (five lobes).

Amplitude coefficient. $c_5 = \sqrt{2}$ (Theorem 1).

Hadron scale context. $M_5 = 460.4$ MeV lies between the kaon (~ 494 MeV) and the eta meson (~ 548 MeV). We do not claim the cinquefoil corresponds to a specific known particle: the mass spectrum depends on θ_5 (open problem 1 below).

5.5 The structural obstruction for $n \geq 5$

With $c_5 = \sqrt{2}$, the five amplitudes $A_k = 1 + \sqrt{2}\cos(\theta + 2\pi k/5)$ for $k = 0, \ldots, 4$ are never all positive simultaneously (see the valid-domain table in Section 4). The minimum amplitude over all five is always negative, with maximum value -0.144 at $\theta = 0$.

This means:

1. $Q_5 = 2/5$ does not hold as a universal constant (Proposition 1 fails).
2. The step $\sum_k \sqrt{m_k} = \sqrt{M}\sum_k A_k$ is invalid because some $A_k < 0$ requires $\sqrt{m_k} = \sqrt{M}|A_k|$.
3. The self-consistency condition $5\theta_5 = Q_5 = 2/5$ cannot be applied without first establishing the correct Q_5.

5.6 The modified Koide ratio for $n \geq 5$

When some $A_k < 0$, define $\tilde{Q}_n(\theta)$ using $|A_k|$:

$$\tilde{Q}_n(\theta) \equiv \frac{\sum_k A_k^2}{\left(\sum_k |A_k|\right)^2}. \tag{9}$$

For $\mathbb{Z}_3$ in its valid domain: $\tilde{Q}_3 = Q_3 = 2/3$ (constant, since all $A_k > 0$ there). For $\mathbb{Z}_5$: $\tilde{Q}_5(\theta)$ varies from 0.294 to 0.326, far from the putative value $2/5 = 0.400$.

The function $\tilde{Q}_5(\theta)$ is not constant. This means: no analogue of the Koide self-consistency argument from Paper V applies to $\mathbb{Z}_5$ without additional structure. A derivation of θ_5 requires either: (a) a physical principle beyond the $\mathbb{Z}_n$ algebraic structure alone, or (b) a modified Brannen parametrisation for the $T(2,5)$ family that avoids negative amplitudes.

6 The CDFT Torus Knot Picture

The torus knot hierarchy $T(2,n)$ for odd n provides a candidate ordered sequence of CDFT particle families. Table 1 lists the energy scales.

The key structural divide is at $n = 3$:

- $n = 3$ (trefoil, leptons): $c_3 = \sqrt{2}$, valid domain exists, $Q_3 = 2/3$ constant, self-consistency closes the internal system.
- $n \geq 5$: $c_n = \sqrt{2}$, *no* valid domain, $Q_n \neq 2/n$ (as a constant). Requires new approach for mass spectrum.

The energy scale hierarchy is fixed once the model assumptions are accepted. The mass spectra of $n \geq 5$ families are open.

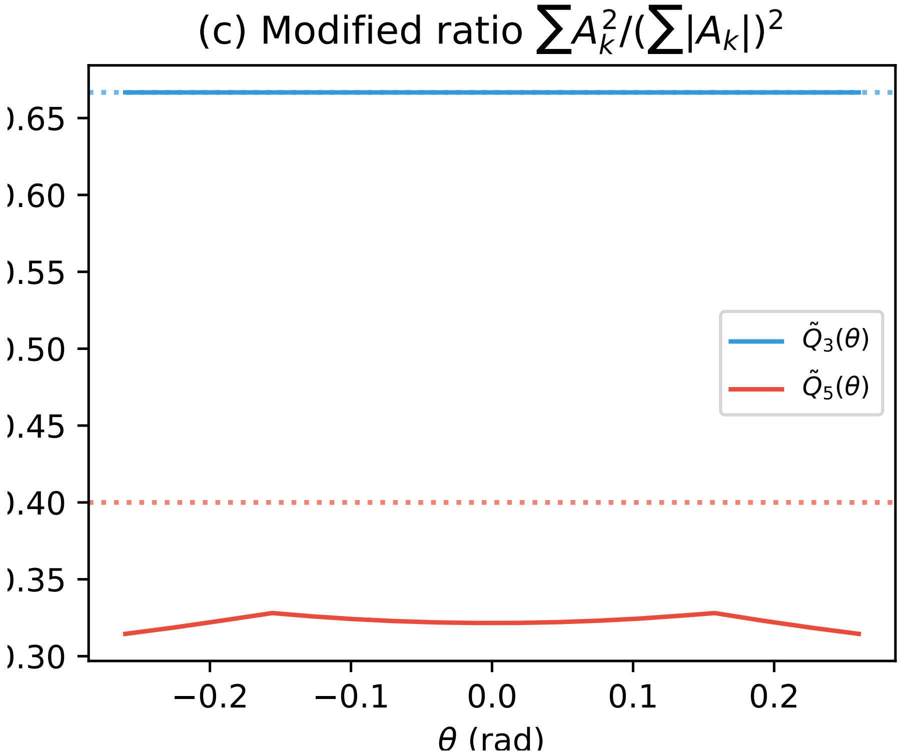

Figure 3: The modified ratio $\tilde{Q}_n(\theta)$ for $n = 3$ and $n = 5$. While $\tilde{Q}_3$ is a strict constant $(2/3)$, $\tilde{Q}_5$ varies and does not meet the $2/5$ target.

7 Open Problems

1. **The $T(2,5)$ chirality phase θ_5.** The cinquefoil has $\mathbb{Z}_5$ symmetry and five modes. Determining θ_5 requires either: (a) a physical principle that selects a preferred orientation of the $\mathbb{Z}_5$ oscillation in the vacuum background, or (b) a modified amplitude form for $T(2,5)$ that ensures all $A_k \geq 0$.

2. **The $T(2,5)$ mass spectrum.** With θ_5 known, the five mode masses are $m_k^{(5)} = M_5|A_k^{(5)}|^2$. Identification of these five modes with known particles (at the ~ 460 MeV scale: kaons, etas, or new states) depends on whether the cinquefoil modes are confined (quark-like) or free (lepton-like).

3. **Why is $\mathbb{Z}_3$ exactly solvable?** Theorem 2 identifies the structural reason ($n = 3$ uniquely has a valid domain with $c_n = \sqrt{2}$), but a deeper question remains: why does the CDFT vacuum select a ground state where the first stable knotted excitation ($T(2,3)$) is exactly the one where the Koide self-consistency is possible?

4. **The general n-family chirality equilibrium.** Paper III derived $3\theta + \phi_c = \pi$ from the $\mathbb{Z}_3$ chirality interaction $\lambda_c \cos(3\theta + \phi_c)$. The $\mathbb{Z}_5$ analogue would be $5\theta_5 + \phi_{c,5} = \pi$ (or some other value). Whether this generalises directly and what determines $\phi_{c,5}$ is an open problem.

Table 2: Torus knot $T(2, n)$ topological classification.

n	Topological type	Persistent?	CDFT family
1	Unknot	No	(none)
2	Hopf link	No	(none)
3	Trefoil knot	**Yes**	Leptons (Papers I–V)
4	Solomon link	No	(none)
5	Cinquefoil knot	**Yes**	This paper
7	Heptafoil knot	**Yes**	Future
$2k$	k-component link	No	(none)
$2k+1$	Torus knot	**Yes**	Predicted

8 Conclusion

This paper extends the CDFT framework beyond the lepton sector by studying the general $\mathbb{Z}_n$ structure of torus knot families $T(2, n)$.

Theorem 1 proves that $c_n = \sqrt{2}$ for all families, from the maximum-entropy self-consistency condition $\mathrm{Var}(A_k) = \overline{A_k}^2$. This is a universal property of the CDFT vacuum that applies across all particle families.

Theorem 2 proves that $\mathbb{Z}_3$ is the unique family where the Koide invariant $Q_n = 2/n$ is constant on a connected valid domain. For $n \geq 5$, the amplitude $A_k = 1 + \sqrt{2}\cos(\cdot)$ is everywhere negative for at least one mode, so the Koide self-consistency of Paper V does not extend. This explains structurally why the lepton sector is exactly solvable and the cinquefoil sector requires new ingredients.

The Faddeev–Niemi energy hierarchy (4) gives the first-principles energy scales of all torus knot families (Table 1), with $M_5 = 460.4$ MeV the firm prediction for the cinquefoil family.

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A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

References

[1] Steve Bico Mujjabi, MD. Vortex stability in a constrained transport vacuum and the origin of the fine-structure constant. *Preprint*, 2026. Paper I of the CDFT series. Derives $\alpha = 1/137.036$ as the aspect ratio of a stable vortex ring in the CDFT vacuum.

[2] Steve Bico Mujjabi, MD. Lepton masses from $\mathbb{Z}_3$-symmetric vortex modes and the origin of the koide formula. *Preprint*, 2026. Paper II of the CDFT series. Proves Koide formula from Z3 symmetry; identifies three charged leptons as trefoil vortex modes.

[3] Steve Bico Mujjabi, MD. Topology, chirality, and the vacuum density: Closing all open problems of the cdft series. *Preprint*, 2026. Paper III of the CDFT series. Closes all four open problems from Papers I–II; derives the chirality equation $3\theta + \phi_c = \pi$; reduces all unknowns to the vacuum density ρ_0.

[4] Steve Bico Mujjabi, MD. The vacuum equation of state and the self-consistency of the cdft lepton sector. *Preprint*, 2026. Paper IV of the CDFT series. Derives the master self-consistency equation for ρ_0; predicts $M_5 = 460.4$ MeV for the cinquefoil $T(2, 5)$ family via Faddeev–Niemi scaling.

[5] Steve Bico Mujjabi, MD. Zero parameters: Deriving the vacuum chirality phase from the koide invariant. *Preprint*, 2026. Paper V of the CDFT series. Derives $\phi_c = \pi - 2/3$ from Z3 self-consistency; gives $\theta = 2/9$ rad; predicts m_μ and m_τ with one input (m_e), zero free parameters.

[6] Carl A. Brannen. Spin path integrals and generations. Foundations of Physics 40, 1681–1699, 2010. Introduces the parametrisation $m_k = M(1 + \sqrt{2}\cos(\theta + 2\pi k/3))^2$ ($k = 0, 1, 2$) and proves it automatically satisfies the Koide formula.

[7] Ludwig Faddeev and Antti J. Niemi. Stable knot-like structures in classical field theory. *Nature*, 387:58–61, 1997.

[8] Peter J. Mohr, David B. Newell, Barry N. Taylor, and Eite Tiesinga. CODATA recommended values of the fundamental physical constants: 2022. *Reviews of Modern Physics*, 97:025002, 2025. $\alpha^{-1} = 137.035\,999\,177(21)$; NIST 2022 CODATA adjustment.

[9] Richard A. Battye and Paul M. Sutcliffe. Knots as stable soliton solutions in a three-dimensional classical field theory. *Physical Review Letters*, 81:4798–4801, 1998. Numerical construction of stable knotted solitons; mass–energy spectrum of torus knot configurations.

[10] Dale Rolfsen. *Knots and Links*. Publish or Perish, 1976. Standard reference on torus knot classification $T(p, q)$, knot invariants, and linking numbers.

[11] Yoshio Koide. New view of quark and lepton mass hierarchy. *Physical Review D*, 28:252–254, 1983. First statement of the charged-lepton mass formula $(m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$; predicted m_τ before precision measurement.